WASP-43b

R-0493

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_230404` · created 2026-07-20T10:42:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460038.7665 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.03
Transit depth (Rp/Rs)^2	1.93 +/- 0.83 [%]
Orbital Inclination (inc)	81.18 +/- 1.99
Airmass coefficient 1 (a1)	0.952 +/- 0.015
Airmass coefficient 2 (a2)	0.031 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#1 - [540, 245]
Optimal Method	PSF photometry
Transit Duration (day)	0.0452 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.03 vs 0.1594 (deviation 0.68σ)
Duration fit vs published	0.0452 d vs 0.0512 d (deviation 0.67σ)
Mid-time O–C	-3.5 min
Depth SNR	4.6
Residual scatter	1.6 %

Light curve

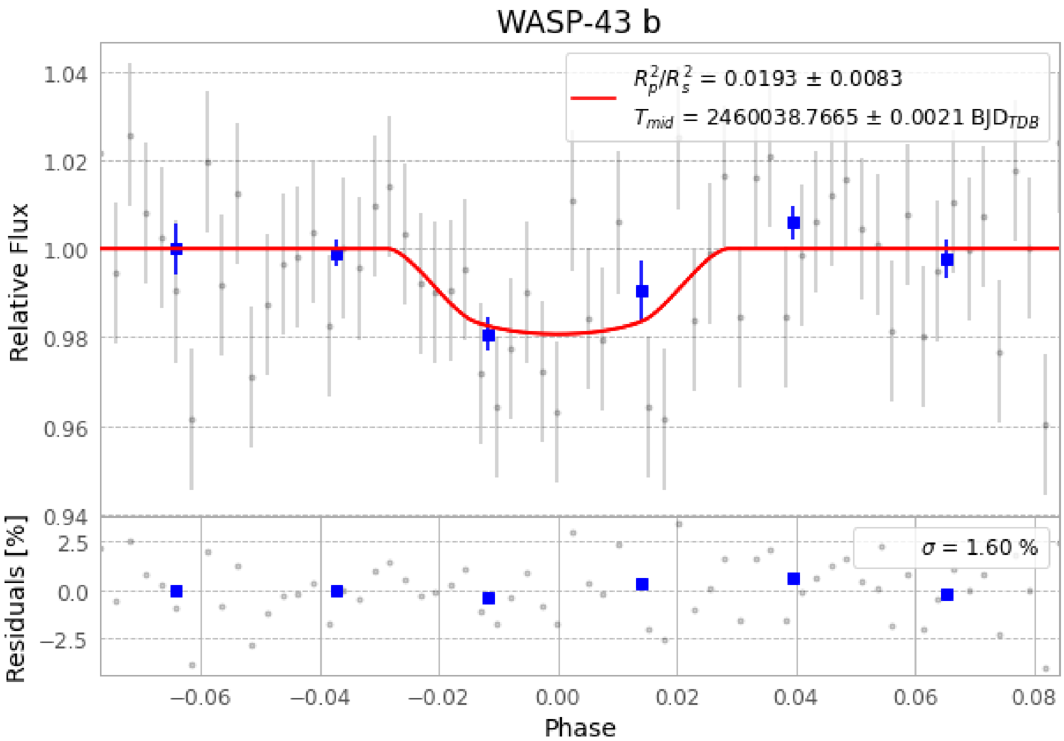

Figure 1 — FinalLightCurve_WASP-43 b_2023-04-03.png (SHA-256 a73ae8b542a4095e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [453, 199], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 cde62dc3767cca89…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

64 FITS frames (42 MB), `WASP-432304040445 15.FITS` → `WASP-432304040754 15.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230404.log (fba9b3c40ca1…), FinalLightCurve_WASP-43 b_2023-04-03.png (a73ae8b542a4…), FinalLightCurve_WASP-43 b_2023-04-03.csv (d2579ccd021e…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)